

SIMATS ENGINEERING

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Name: Data Warehousing and Data Mining

Course Code: CSA16

Course Outcome: CO1 – Design data warehouse architectures with appropriate dimensional models for effective decision support systems.

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Question 1

Given:

A data warehouse consists of three dimensions: **Time, Doctor, and Patient**, with two measures: **Count** and **Charge**.

(a) Three popular data warehouse schemas

The three commonly used schemas for modeling data warehouses are:

1. **Star Schema**
2. **Snowflake Schema**
3. **Fact Constellation Schema**

Star Schema:

A central fact table is directly connected to several denormalized dimension tables. It is simple and provides efficient query performance.

Snowflake Schema:

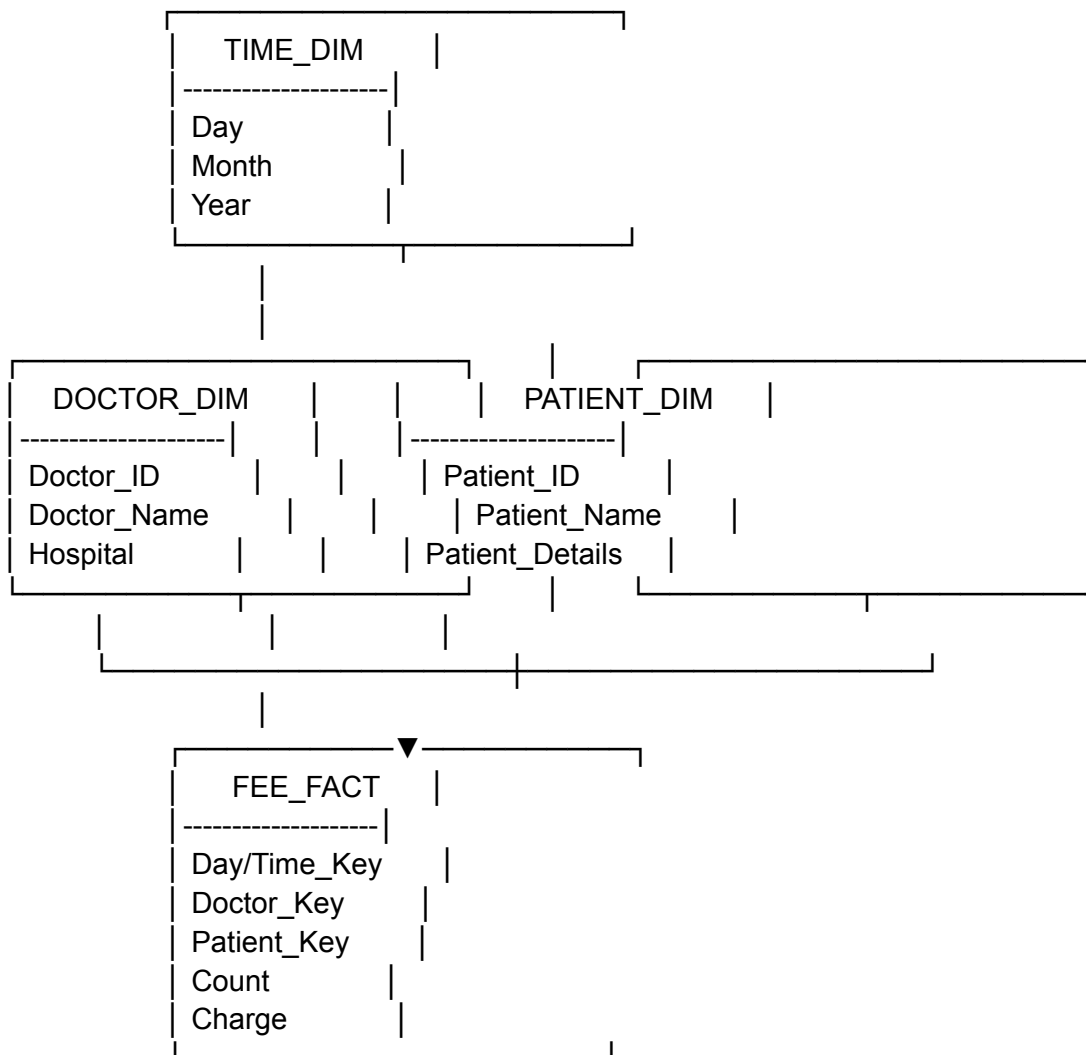
An extension of the star schema in which dimension tables are normalized into multiple related tables. It reduces redundancy but may require more joins.

Fact Constellation Schema:

Also called a galaxy schema. It contains multiple fact tables that share common dimension tables.

(b) Schema diagram

A **Star Schema** can be used for this data warehouse.



Here, **FEE_FACT** is the central fact table. The dimensions provide descriptive information, while **Count** and **Charge** are the measures.

(c) SQL query

Given the relational schema:

`fee(day, month, year, doctor, hospital, patient, count, charge)`

To obtain the list containing the doctor, hospital, patient, count, and charge for a particular period:

```
SELECT
    day,
    month,
    year,
    doctor,
    hospital,
    patient,
    count,
    charge
FROM fee;
```

If the requirement is to obtain aggregated information for each doctor and patient, the query can be written as:

```
SELECT
    doctor,
    hospital,
    patient,
    SUM(count) AS total_visits,
    SUM(charge) AS total_charge
FROM fee
GROUP BY doctor, hospital, patient;
```

Thus, the star schema provides a simple dimensional representation and supports efficient analytical queries.

Question 2

A data warehouse for **Big University** consists of four dimensions:

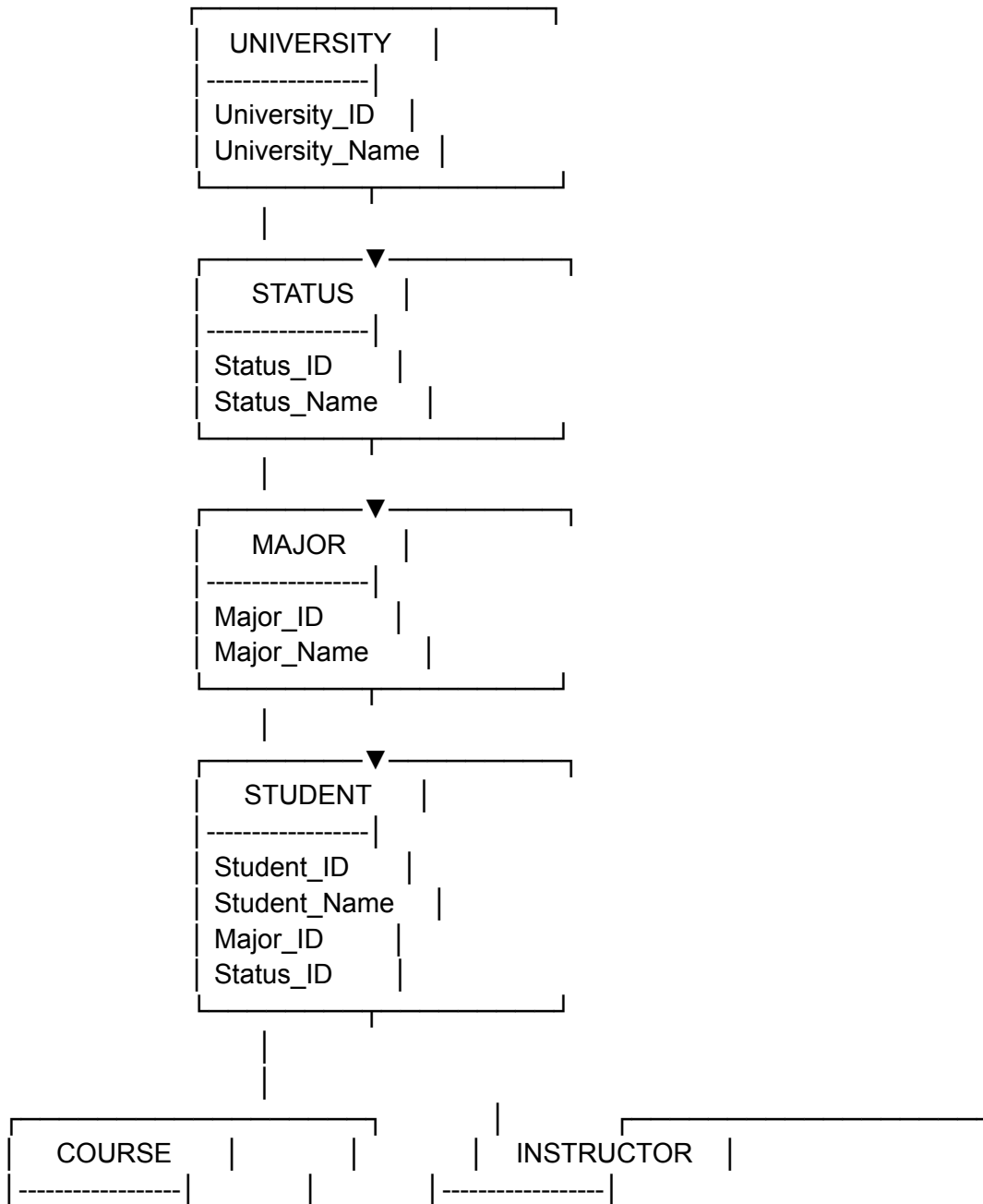
- Student
- Course
- Semester
- Instructor

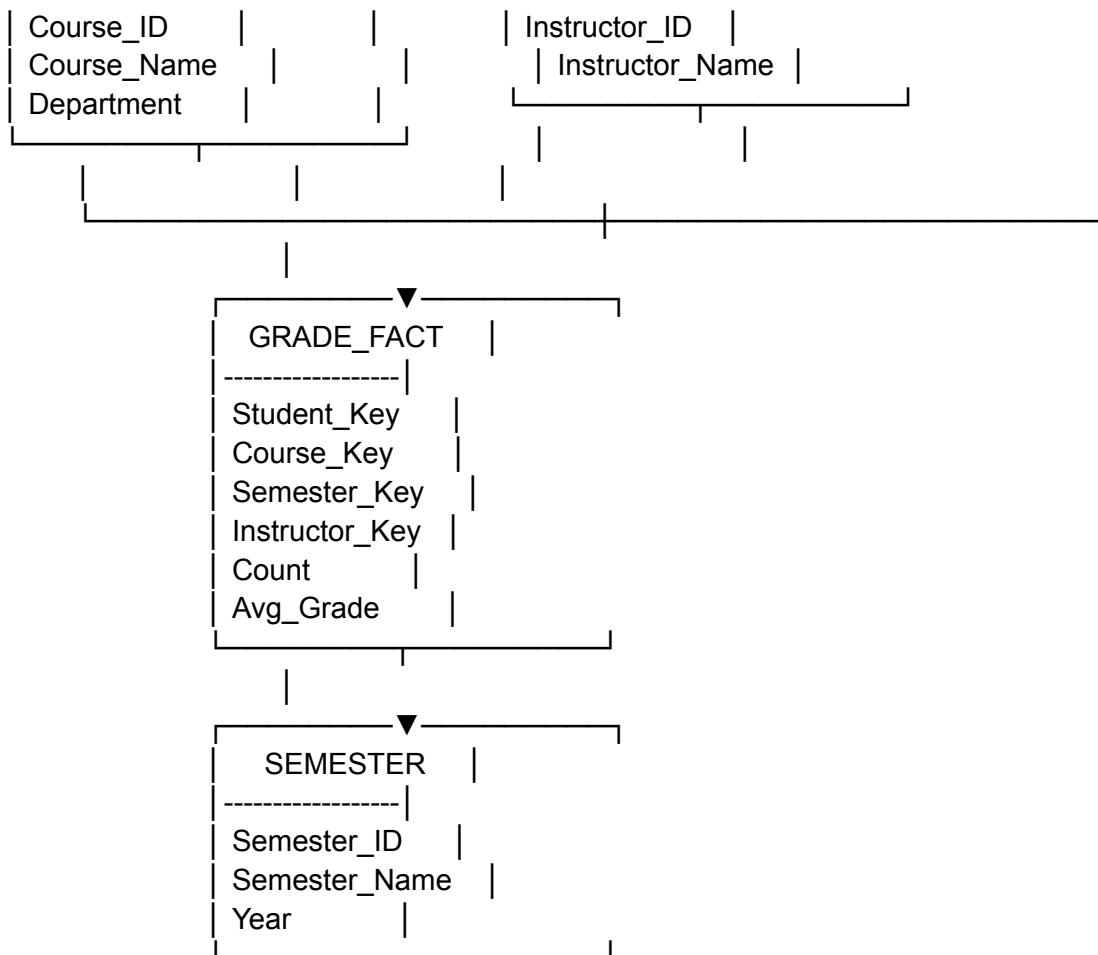
Measures:

- Count
- Average Grade

At the lowest level, **avg grade** stores the actual grade obtained by a student. At higher levels, it stores the average grade.

(a) Snowflake Schema Diagram





This is a snowflake schema because the **Student dimension is normalized into multiple related tables**, reducing redundancy.

(b) OLAP operations

The base cuboid is:

[Student, Course, Semester, Instructor]

The objective is to list the **average grade of CS courses for each Big University student**.

The required OLAP operations are:

Step 1 – Roll-up Instructor

Roll up the **Instructor** dimension to the **ALL** level because the instructor is not required in the final result.

Instructor → ALL

Step 2 – Roll-up Semester

Roll up the **Semester** dimension to **ALL** if the result should contain the overall average grade across semesters.

Semester → ALL

Step 3 – Slice/Dice Course

Select only courses belonging to the **Computer Science (CS)** department.

Course.Department = 'CS'

Step 4 – Group by Student

Keep the **Student** dimension at the required level.

The resulting cuboid is approximately:

[Student, CS Course, ALL Semester, ALL Instructor]

Step 5 – Calculate average grade

The **avg_grade** measure is aggregated using the **average operation**.

The final output is:

Student_ID | Student_Name | Average_CS_Grade

Therefore, the main operations are **roll-up, slice/dice, and aggregation**.

(c) Number of cuboids

Each dimension has **5 levels**, including the **ALL** level.

For a data cube with four dimensions:

Number of cuboids = $5 \times 5 \times 5 \times 5$

$$= 5^4$$
$$= 625$$

Therefore:

Total number of cuboids = 625

This includes both the **base cuboid** and the **apex cuboid**.

Question 3

Data Warehouse Design for a Regional Weather Bureau

The weather bureau collects data from approximately **1,000 probes** distributed across land and ocean locations. Each probe records:

- Air pressure
- Temperature
- Precipitation

The readings are collected **every hour** and stored for more than **10 years**.

A suitable solution is a **multidimensional data warehouse using a star schema**.

1. Fact Table

The central fact table can be called:

WEATHER_FACT

It contains the measured weather information.

WEATHER_FACT

Time_Key
Location_Key
Probe_Key
Temperature
Air_Pressure
Precipitation

Reading_Count

The measures are:

- Temperature
- Air Pressure
- Precipitation
- Reading Count

2. Time Dimension

TIME_DIM

Time_Key

Hour

Day

Month

Quarter

Year

The hierarchy is:

Hour → Day → Month → Quarter → Year → ALL

This allows analysis at hourly, daily, monthly, quarterly, and yearly levels.

3. Location Dimension

LOCATION_DIM

Location_Key

Location_Name

Region

Country

Land_Ocean

Latitude

Longitude

A geographic hierarchy can be:

Location → Region → Country → ALL

Latitude and longitude can also be used for spatial analysis.

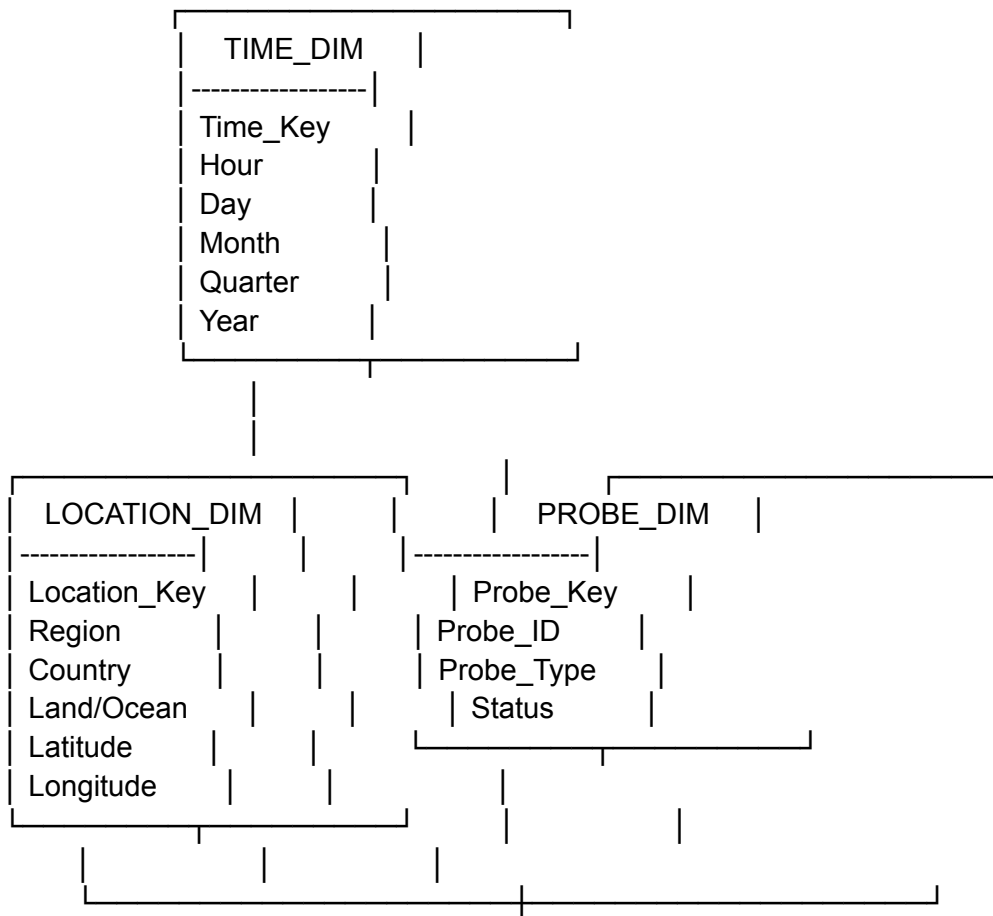
4. Probe Dimension

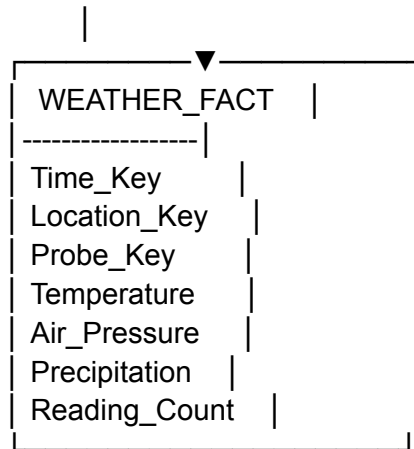
PROBE_DIM

Probe_Key
Probe_ID
Probe_Type
Installation_Date
Location_Key
Status

This dimension helps analyze readings from individual probes.

5. Star Schema





6. ETL Process

The warehouse can use the following ETL process:

Weather Probes



Data Collection



Data Cleaning



Transformation



Central Data Warehouse



OLAP / Data Analysis



Reports & Weather Patterns

The data should be cleaned to handle missing readings, duplicate records, sensor errors, and invalid measurements.

7. OLAP Operations

The warehouse supports:

Roll-up

Example:

Hour → Day → Month → Year

This can identify long-term temperature trends.

Drill-down

Example:

Year → Month → Day → Hour

This allows detailed investigation of unusual weather events.

Slice

Select a single dimension value, such as:

Year = 2026

Dice

Select multiple conditions, such as:

Year = 2026

Region = Coastal

Temperature > 35°C

8. Weather Pattern Analysis

The warehouse can help identify:

- Average temperature by month
- Annual rainfall patterns
- High-pressure and low-pressure regions
- Seasonal weather trends
- Coastal versus inland temperature differences
- Extreme weather events
- Long-term climate patterns
- Locations receiving maximum precipitation

Because the warehouse stores more than 10 years of hourly data, historical trends can also be analyzed efficiently.

9. Benefits

The proposed warehouse provides:

1. Efficient multidimensional querying.
2. Fast OLAP operations.
3. Historical weather analysis.
4. Spatial and temporal pattern detection.
5. Easy generation of reports and dashboards.
6. Support for identifying abnormal weather conditions.
7. Better decision-making for weather forecasting and environmental monitoring.

Thus, a **star-schema weather data warehouse with Time, Location, and Probe dimensions** provides an effective solution for analyzing large-scale historical weather data.

Question 4

A data cube **C** has **n** dimensions, and each dimension has exactly **p** distinct values in the base cuboid. There are no concept hierarchies associated with the dimensions.

(a) Maximum number of cells in the base cuboid

Each of the **n** dimensions contains **p** distinct values.

Therefore, the maximum number of cells is:

$$p \times p \times \dots \times p \text{ } p \text{ times } p \text{ } \times \text{ } p \text{ } \times \text{ } \dots \times \text{ } p \text{ } \times \text{ } p$$

for **n** dimensions.

Hence:

$$p^n$$

Answer: Maximum base-cuboid cells = p^n

(b) Minimum number of cells in the base cuboid

If each dimension has p distinct values, the base cuboid must contain every possible combination of these values.

Therefore, even in the minimum case:

$$p^n$$

Answer: Minimum base-cuboid cells = p^n

This is because the number of distinct values in every dimension is fixed at p .

(c) Maximum number of cells in the complete data cube

Since there are no concept hierarchies, each dimension can either:

1. Remain at its detailed level, or
2. Be aggregated to **ALL**.

Therefore, each dimension has $p + 1$ possible values in the complete cube.

For n dimensions:

$$(p+1)^n$$

Thus, the maximum number of cells, including base cells and aggregate cells, is:

$$(p+1)^n$$

(d) Minimum number of cells in the complete data cube

The minimum number of cells occurs when only the required aggregate combinations are populated while maintaining the base-level cells.

The base cuboid contains:

$$p^n$$

cells.

Each dimension also produces aggregate cells. For the minimum complete cube, the aggregate cells can be shared as much as possible.

The minimum number of cells is therefore:

$$p^n + np + 1$$

where:

- p^n = base-level cells
- np = one-dimensional aggregate cells across all n dimensions
- 1 = apex cuboid cell

Therefore:

$$\text{Minimum cells} = p^n + np + 1$$

Final Answers – Quick Reference

Question	Answer
1(a)	Star, Snowflake, Fact Constellation
1(b)	Star schema with Time, Doctor, Patient dimensions and Fee fact
1(c)	SQL SELECT / aggregation using the fee relation
2(a)	Snowflake schema with Student, Course, Semester, Instructor
2(b)	Roll-up Instructor and Semester, slice/dice CS courses, aggregate by Student
2(c)	$54 = 6255^4 = \mathbf{625}$ cuboids
3	Star schema with Weather Fact, Time, Location, and Probe dimensions
4(a)	p^n
4(b)	p^n
4(c)	$(p+1)^n$
4(d)	$p^n + np + 1$

Conclusion

Data warehousing provides a structured environment for storing large volumes of historical data and performing efficient analytical operations. Star and snowflake schemas organize facts and dimensions for decision support, while OLAP operations such as **roll-up, drill-down, slice, and dice** allow users to analyze information at different levels of detail. The weather bureau example demonstrates how multidimensional modeling can support temporal, geographic, and sensor-based analysis. Understanding the mathematical structure of data cubes further helps in estimating storage requirements and designing scalable data warehouse systems.